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|ITESM Campus Chihuahua

Paper #12 · Oral

Forecasting Weekly Depression Incidence in Mexico:
A Multi-Model Framework with Auditable Per-Series Model Selection

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Independent research; affiliations do not imply institutional endorsement.

When historical scores meet recorded 2026 forecasts

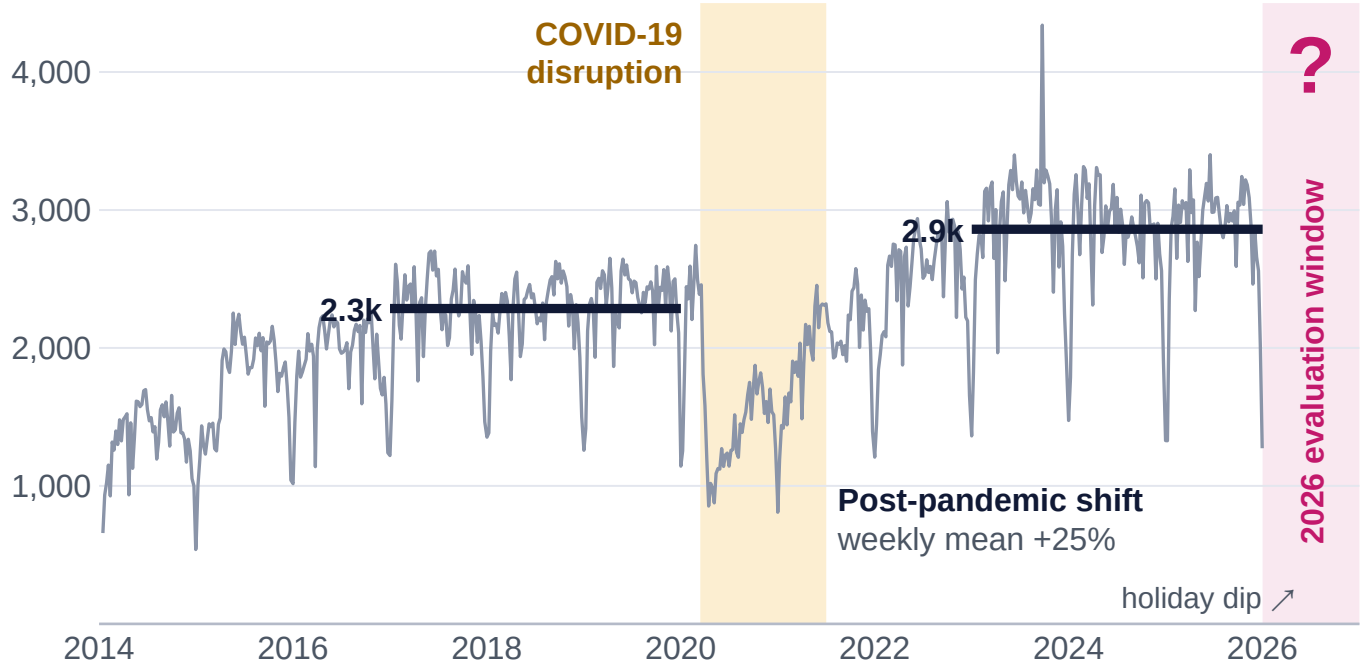


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01 THE PROBLEM

Depression is counted weekly, but only in hindsight

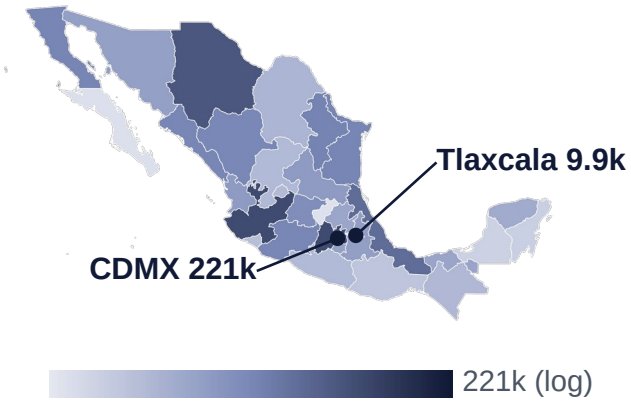
Weekly reported cases, national ICD-10 F32, SINAVE 2014–2025



~150k
cases per year
2023–2025

74%
are women
annual totals by sex

~22× **total reported cases, 2014–2025**



Source: SINAVE bulletins [1]; reported counts, not prevalence. Lines: weekly means 2017–19, 2023–25. Curve omits 2014-W02 and 2016-W20 (flagged partial).

02 THE FRAMEWORK

Four candidates per series, one auditable rule

SINAVE bulletins

Weekly ICD-10 F32 counts, 2014–2025 (626 weeks)

111 targets: 32 states, 4 regions, national; general, women, men

99 reported directly

Four candidates

- **DeepAR** [3] probabilistic recurrent net
- **Prophet** [2] trend + seasonality
- **Ensemble** [4] Prophet + XGBoost blend
- **Stacking** [5, 6] Prophet, ETS, LightGBM

Deterministic selection rule

1. **Lowest historical sMAPE**
2. **Within 5% relative: lowest MASE**
3. **MASE tie: lowest RMSE**

Recorded forecast

52 weeks ahead

Audit log

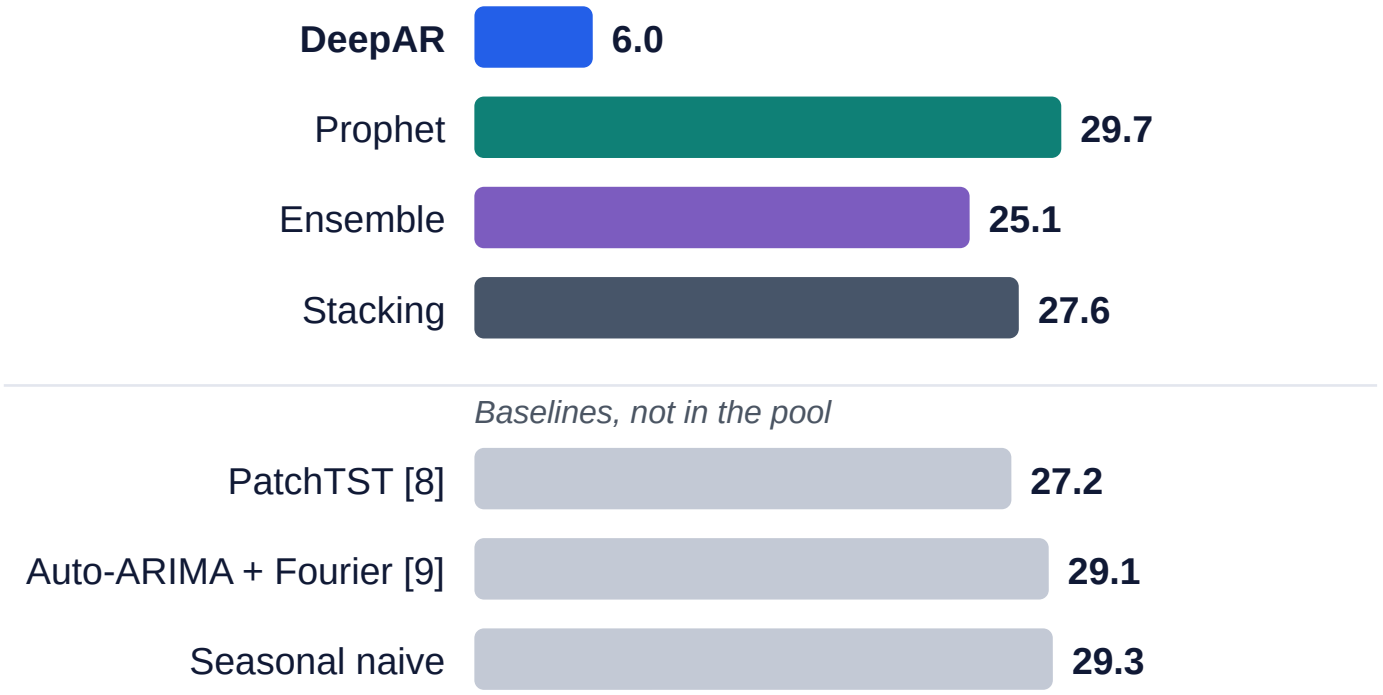
which model, on what evidence

Selection evidence: per-model historical scores (paper Table 1); protocols differ by model. MASE as in [7]. The sparse fallback never triggered (0 of 111).

03 SELECTION EVIDENCE

Historical selection favored DeepAR

Median historical sMAPE (%) Paper Table 1 · 111 targets · lower is better



Series the rule assigns to DeepAR

107 / 111

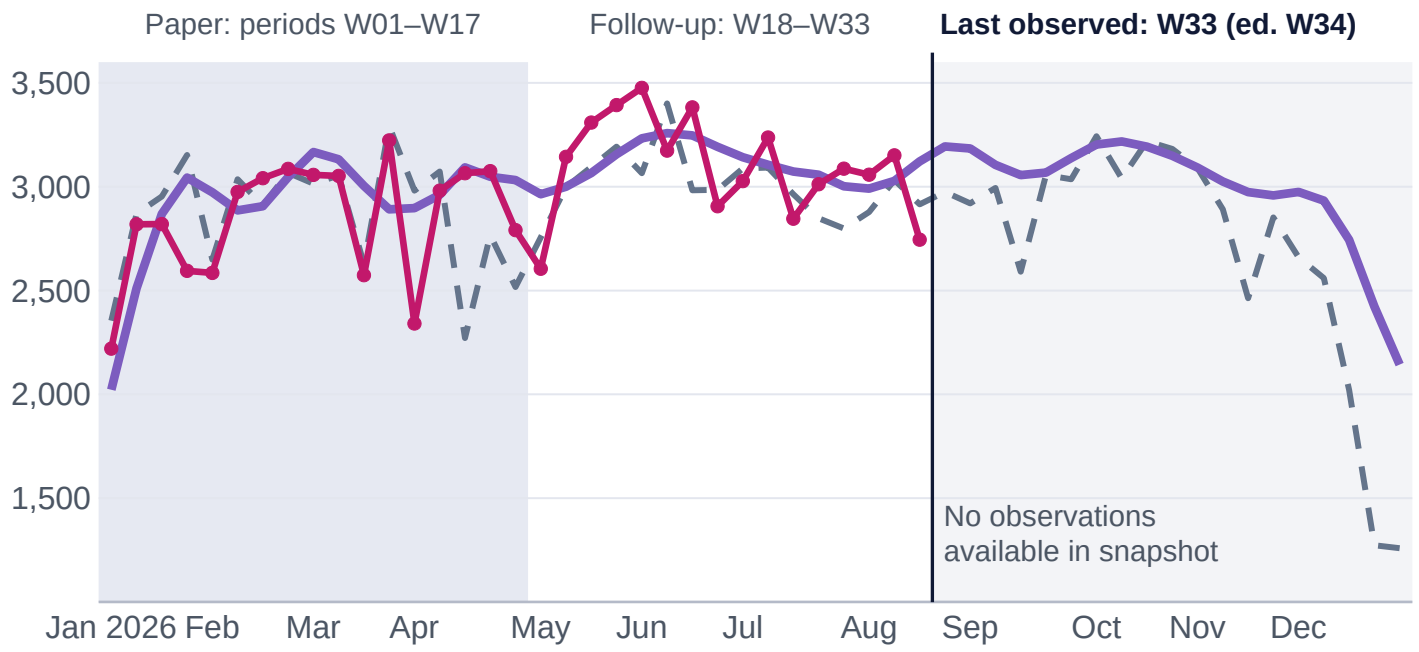
96.4%, including every state series

Reading the 6.0: in the historical code, DeepAR's state scores reflect training fit, not forecast error.

DeepAR's 6.0% historical sMAPE is the median over all 111 targets. Historical selection scores are not a guarantee of 2026 accuracy.

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Recorded trajectories and available follow-up



Paper sMAPE (%)
editions W02–W18 · **Lower: naive**

Ensemble	**7.40%**
52-week naive	**7.33%**

Follow-up sMAPE (%)
editions W19–W34 · **Lower: Ensemble**

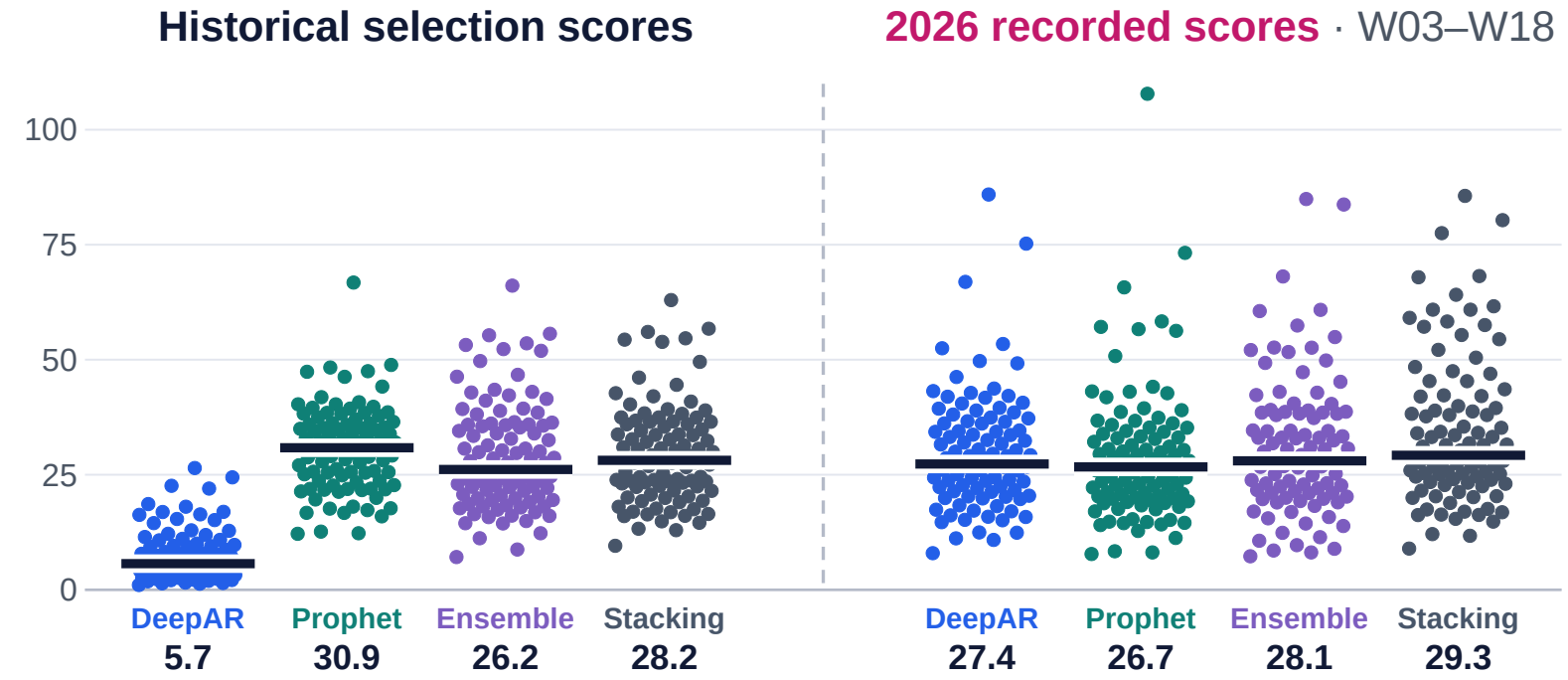
Ensemble	**5.91%**
52-week naive	**6.24%**

Recorded forecasts · retrospective evaluation

sMAPE, lower is better. Cumulative deviation, Ensemble / naive: paper +2.45% / +0.6%; follow-up +0.17% / −2.95%. Snapshot: edition W34. Descriptive.

Historical and 2026 model rankings differ

Per-series sMAPE (%) 99 series, one dot each · bar = median · lower is better



Historical score gap

≈20 pts

Range of 2026 medians

< 3 pts

No model is best everywhere

35 | 27 | 19 | 18

series won in 2026: Prophet, DeepAR, Ensemble, Stacking

Historical protocols differ by model. Post hoc and exploratory: targets overlap; paired tests in backup B2.

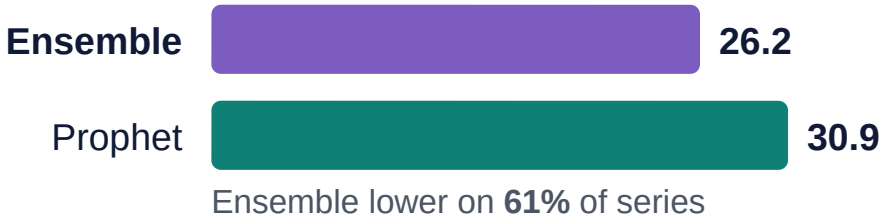
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06 WITHOUT DEEPAR

Excluding DeepAR, the two rankings differ

Median sMAPE (%) Ensemble vs. Prophet · same 99 series · lower is better

Historical held-out windows



Ensemble lower on **61%** of series

2026 recorded scores W03–W18



Prophet lower on **60%** of series

Same lowest-error model, historical vs. 2026

Prophet, Ensemble, Stacking · 99 series

34% **observed**
34 of 99 series

31% **reference**
from each model's win frequencies

Descriptive reference, not a formal chance test.

Additional analysis beyond the paper. Point rankings; post hoc and exploratory. Averaging each state's three targets weakens the reversal (backup B2).

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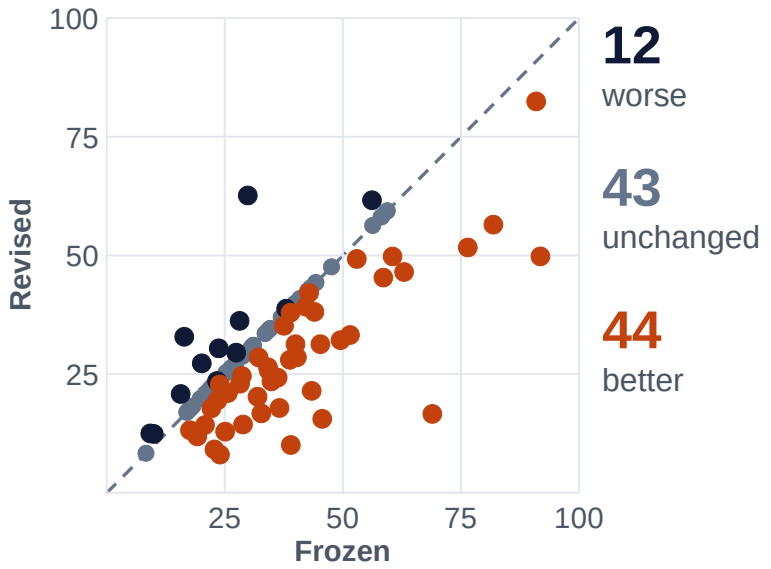
07 REVISION REPLAY

Revision lowered the median in a seven-week replay

W03–W11 re-select: same rule, prior weeks only

W12–W18 evaluation: excluded from reselection

Each dot is one series sMAPE, W12–W18



12 worse

43 unchanged

44 better

Policies median sMAPE, W12–W18

Frozen assignment	32.7
Global re-selection	32.7
Per-series revision	**28.5**
Comparators	
Prophet only	27.6
Pool without DeepAR	26.4

−4.2 pts difference of policy medians

Reading

In this seven-week replay, revision reduced the median from 32.7% to 28.5%.

Simple alternatives had lower medians.

Revised vs. frozen: 44 better, 12 worse, 43 equal. Overall medians were lower at all nine splits (W06–W14); at W11 they were lower in each stratum. Exploratory.

TAKEAWAYS

Selection is a signal, not a guarantee

An auditable per-series selection framework, evaluated on recorded 2026 forecasts.

1

Historical and 2026 rankings differ

Historical selection favored DeepAR. Prophet had the lowest median error in the recorded panel.

2

Revision lowered the replay median

Seven-week replay: revision 32.7 → 28.5; pool without DeepAR 26.4.

3

The year so far

16-week follow-up: national Ensemble 5.91 vs naive 6.24; archived panel median 31.0 vs 27.0.

THANK YOU

Questions?

Selection is a signal, not a guarantee.

Forecasting Weekly Depression Incidence in Mexico: A Multi-Model Framework with Auditable Per-Series Model Selection

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github.com/EpiForecastMX/per-series-model-selection2026

epiforecast.mx/micai2026
follow-up, data and slides

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SOURCES

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This work. Rebull-Saucedo, J.A., Pérez-Nava, J.C., Sánchez-Salazar, L.G., Barceló-Alonso, G., Pérez-Hernández, R.: Forecasting weekly depression incidence in Mexico: a multi-model framework with auditable per-series model selection. In: Advances in Soft Computing, MICAI 2026, LNCS 17149, Springer (to appear).

The selection rule, exactly

Per series, with recorded historical sMAPE, MASE, and RMSE per model:

1. If the second-lowest sMAPE exceeds 1.05 times the lowest, choose the lowest.
2. Otherwise, among models with sMAPE ≤ 1.05 × best, pick the lowest MASE.
3. If MASE is tied, pick the lowest RMSE.

\+ Fewer than 5 cases in the 52 weeks before the forecast origin: use the regional model (past data only; never triggered here, 0 of 111).

Each choice and its metrics are stored with the forecast: the audit log.

Selection evidence

Per-model historical backtests (paper Table 1), not one shared protocol.

Prophet, Ensemble, Stacking: held-out windows, with different protocols.

Historical state DeepAR code skips CV, trains on the full series, then scores that fit (commit 60d6ea0f; changed in 9fdd6dca). The exact published runs are not independently linked.

The paper reports a data cutoff of 2025-W53; execution dates are not independently verified.

Statistical tests and uncertainty

Comparison	99 targets: raw / Holm	32 states: nominal
Historical, Ensemble vs. Prophet	0.020 / 0.060	0.15
2026, Prophet vs. Ensemble	0.007 / 0.035	0.06
2026, Prophet vs. DeepAR	0.070 / 0.14	0.24
Revision vs. frozen, split W11	< 0.001 / < 0.001	< 0.001

Exploratory, post hoc; targets overlap and Holm does not remove dependence. Not significant ≠ equivalent. Two-sided Wilcoxon, zsplit; Holm within six model pairs (nine for revision).

National, W03–W18, recomputed

Mean Poisson deviance: Ensemble 25.7, Prophet 32.3, DeepAR 46.0, Stacking 97.8.

Reported in the paper only

Diebold–Mariano tests; the 80% band covered 10 of 17 weeks. Inputs not released.

Limitations

One condition

Depression (F32) only; transfer is untested.

Training-fit evidence

Historical state DeepAR code uses training fit; run provenance is unverified.

Unequal search budget

Prophet had the broadest search; tuning effect untested.

Partial-year evaluation

Paper windows plus 16 follow-up periods; full-year evaluation remains pending.

Provenance

Forecast execution dates not independently verified.

Uncertainty

Paper-reported 80% band coverage: 58.8%; revision does not calibrate it.

Reported incidence

Models learn reporting and disease together.

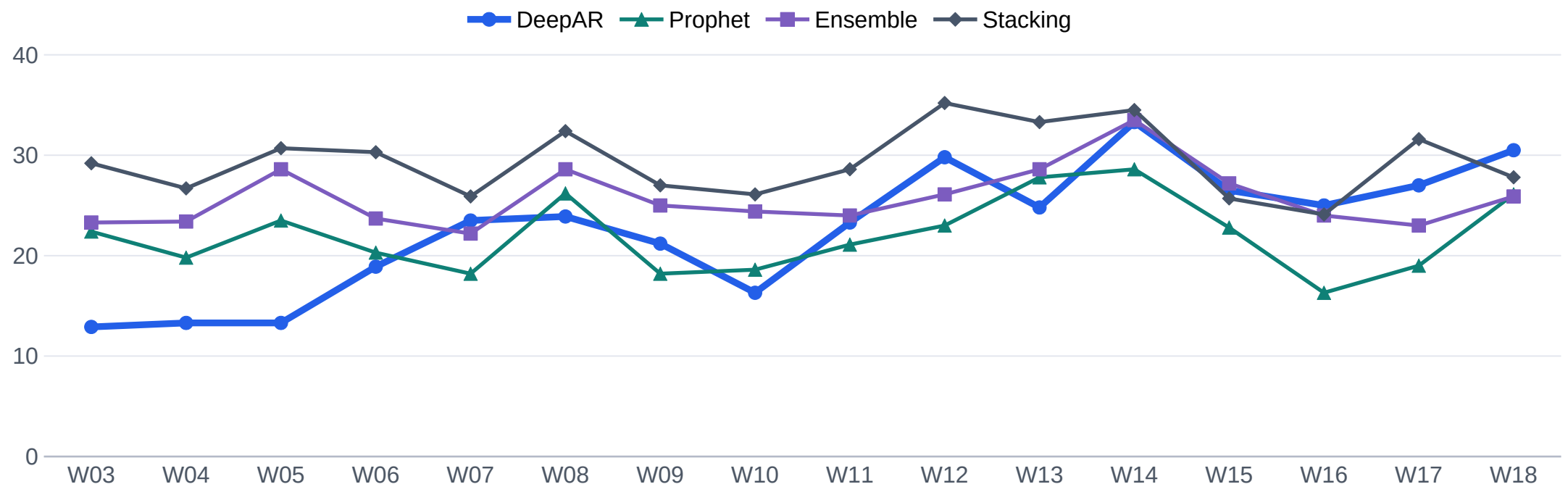
No external validation

Only SINAVE / Mexico; a research prototype.

B4 BACKUP · HORIZON

DeepAR leads the first four editions (W03–W06)

Median per-week sMAPE (%) across the 99 series 2026 bulletin editions · lower is better

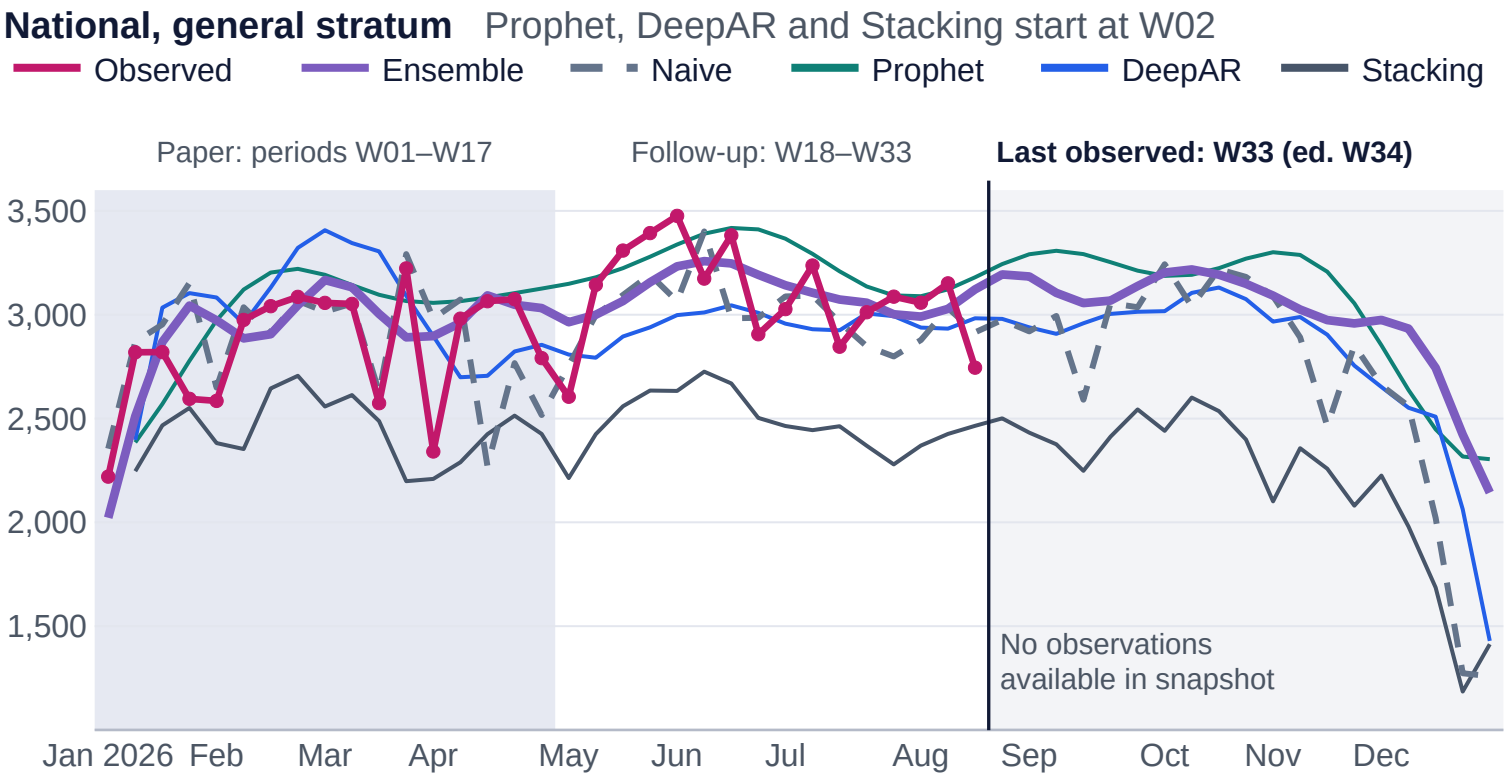


Revision chooses among frozen 52-week paths; horizon effects may contribute to its gain (not isolated). Recomputed from the public files.

B5 BACKUP · FOLLOW-UP

Recorded 2026 paths: all four engines

National, general stratum Prophet, DeepAR and Stacking start at W02



99 targets, editions W19–W34
median sMAPE · targets below naive

Prophet	**23.4**	70/99
Ensemble	**25.2**	71/99
DeepAR	**31.0**	52/99
Stacking	**30.6**	51/99
Archived	**31.0**	55/99
Naive	**27.0**	

Forecast periods mapped by rule; execution origin unverified.

Descriptive; targets overlap. W01 reproduces the paper; legacy mapping is ambiguous. Archived = frozen assignment (96 DeepAR).

B6 BACKUP · BY SEX

Every model against the naive, by window and sex

Median sMAPE (%) and targets beating the naive, of 33 per stratum · lower is better

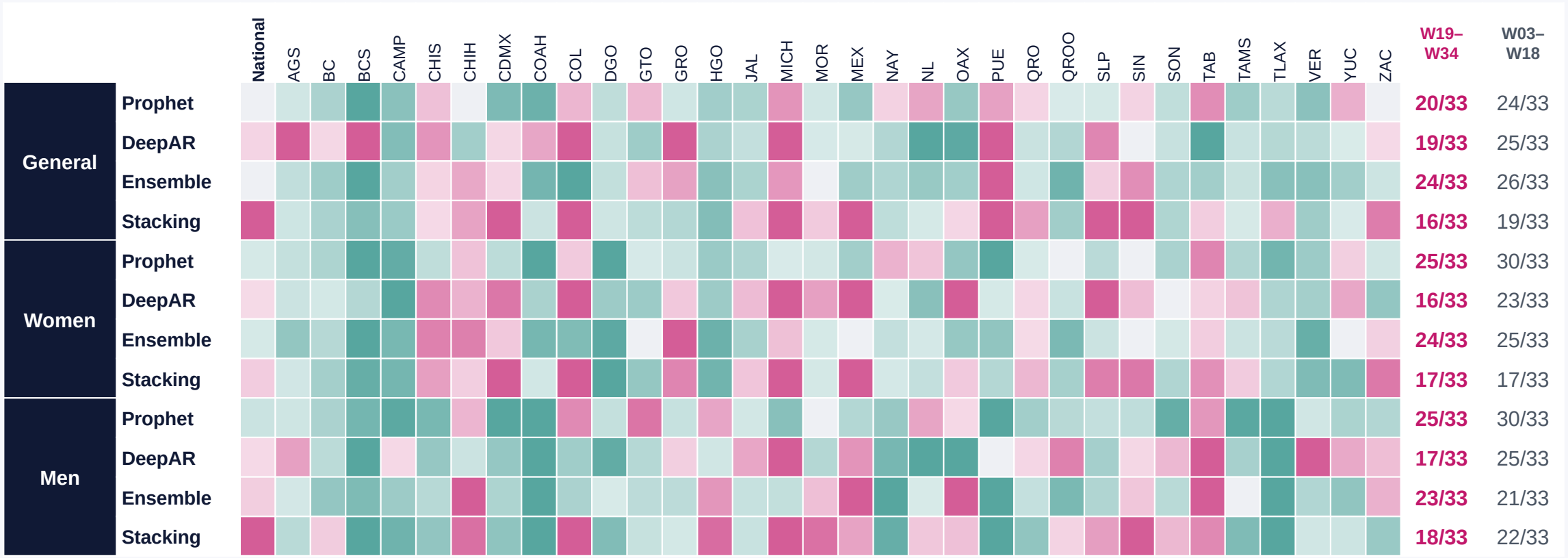
	Paper window · editions W03–W18			Follow-up · editions W19–W34		
Estimator	**General**	**Women**	**Men**	**General**	**Women**	**Men**
Prophet	**23.8 · 24**	24.4 · 30	**30.4 · 30**	**20.1 · 20**	**20.3 · 25**	**32.3 · 25**
DeepAR	25.9 · 25	26.0 · 23	32.4 · 25	22.0 · 19	29.6+ · 16	34.3 · 17
Ensemble	26.7 · 26	**23.8 · 25**	38.0 · 21	22.7+ · 24	24.8 · 24	34.5 · 23
Stacking	29.0 · 19	27.9 · 17	34.3 · 22	25.9+ · 16	30.2+ · 17	34.3 · 18
Archived assignment	25.9 · 26	26.0 · 24	32.4 · 25	22.0 · 20	29.6+ · 17	34.3 · 18
52-week naive	31.2	30.6	40.1	22.5	24.9	36.5

Cell: median sMAPE across 33 targets · targets with lower sMAPE than the naive. + = median above the naive; bold = lowest median. Archived assignment = DeepAR in every state. Descriptive; overlapping targets.

B7 BACKUP · BY STATE

How the year is going: every model against the naive

Follow-up, editions W19–W34 · cell = model sMAPE minus naive sMAPE, one target each



Model minus naive (pts): ≤−15 · −8 · −3 · ±0.5 · +3 · +8 · ≥+15

Teal: lower error than the naive (better). Magenta: higher. Gray: under 0.5 pts, not necessarily a tie. Archived assignment = DeepAR in every state. Descriptive; overlapping targets. Full matrix for download; state-level reading requires zoom.

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